

Model Question Paper-I with effect from 2025

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1BMATC101

First Semester B.E./B.Tech. Degree Examination Differential Calculus and Linear Algebra

TIME: 03Hours

Max.Marks:100

- Note: 1. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**
 2. VTU Formula Hand Book is Permitted
 3. M: Marks, L: Bloom's level, C: Course outcomes

Module-1			M	L	CO
Q 1.	a	With usual notations, prove that $\tan(\phi) = r \frac{d\theta}{dr}$.	6	L2	1
	b	Find the angle between pair of curves, $r \sec^2\left(\frac{\theta}{2}\right) = 2a$, $r \operatorname{cosec}^2\left(\frac{\theta}{2}\right) = 2b$.	7	L2	1
	c	Find the pedal equation of the curve $r^n = a^n \cos n\theta$.	7	L2	1
OR					
Q 2.	a	Find the radius of curvature of the curve $x^3 + y^3 = 3axy$, at the point $\left(\frac{3a}{2}, \frac{3a}{2}\right)$.	6	L2	1
	b	Find the radius of curvature of the parametric curve $x = a(t - \sin t)$, $y = a(1 - \cos t)$.	7	L2	1
	c	Show that ρ varies inversely as r^{n-1} for the curve $r^n = a^n \sin(n\theta)$,	7	L2	1
Module-2					
Q 3.	a	Obtain the series of the function $y = \log(1 + \cos x)$ upto the term contains x^4	6	L2	1
	b	Evaluate (i). $\lim_{x \rightarrow a} \left(\frac{x^x - a^x}{x^a - a^a} \right)$ and (ii). $\lim_{x \rightarrow 0} \frac{\sinh(x) - x}{\sin(x) - x \cos(x)}$.	7	L2	1
	c	If $u = x^3 - 3xy^2 + x + e^x \cos y + 1$ show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$	7	L2	1
OR					
Q 4.	a	If $z = f(x, y)$ where $x = e^u \sin v$ and $y = e^u \cos v$ then prove that $\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2 = e^{2u} \left\{ \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 \right\}$	6	L2	1
	b	If $u = \frac{yz}{x}$, $v = \frac{xz}{y}$, $w = \frac{xy}{z}$ show that $\frac{\partial(u, v, w)}{\partial(x, y, z)} = 4$	7	L2	1
	c	Find the extreme value of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$	7	L2	1
Module-3					
Q 5.	a	Solve the differential equation $\cos x \, dy = y(\sin x - y)dx$	6	L2	1

	b	Solve the differential equation $(x - y)dx - dy = 0$, $y(0) = 2$	7	L2	1
	c	Solve the differential equation $(3x^2y^4 + 2xy)dx + (2x^3y^3 - x^2)dy = 0$	7	L2	1
OR					
Q 6.	a	Find the orthogonal trajectories of $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1$ where λ is the parameter.	6	L2	1
	b	A bacterial population B is known to have a rate growth proportional to B itself. If between noon and 2PM the population triples, at what time no controls being exerted, should B become 100times what it was at noon.	7	L3	1
	c	Radium decomposes at a rate proportional the quantity of radium present. Suppose that it is found that in 25 years approximately 1.1% of a certain quantity of radium has decomposed. Determine approximately how long will it take for half of the original amount of radium to decompose. -	7	L3	1
Module-4					
Q 7.	a	Solve $y^{111} - 3y^{11} - y^1 + 3y = 0$	6	L2	1
	b	Solve $(D^2 - D + 1)y = \sinh x$	7	L2	1
	c	Solve $(D^2 + 5D - 6)y = \sin 4x \sin x$	7	L2	1
OR					
Q 8.	a	Solve $(D^2 + 1)y = \cos x$ by the Method of variation of parameters.	6	L2	1
	b	Solve $x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} 5y = x^2 \cos x (\log x)$	7	L2	1
	c	Construct the governing equation of the free, damped motion of a mass?	7	L2	1
Module-5					
Q 9.	a	Find the rank of the matrix $A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & 7 \end{bmatrix}$	6	L2	2
	b	Test for the consistency of the following system and then solve $x - 2y + 3z = 2$, $3x - y + 4z = 4$, $2x + y - 2z = 5$.	7	L2	2
	c	Solve the following system of equation by Guass elimination method $2x_1 + 2x_2 + x_3 + 2x_4 = 7$, $-x_1 + 2x_2 + x_4 = -2$, $-3x_1 + x_2 + 2x_3 + x_4 = -3$, $-x_1 + 2x_4 = 0$.	7	L3	2
OR					
Q 10.	a	Solve the following system of equations by Gauss seidel method $10x + y + z = 12$, $x + 10y + z = 12$, $x + y + 10z = 12$	6	L3	2
	b	Apply power method to find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} -2 & 2 & 3 \\ 2 & 1 & 6 \\ 3 & 6 & -2 \end{bmatrix}$.	7	L3	2
	c	Diagonalize the matrix $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.	7	L2	2

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1BMATM101

First Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026 Differential Calculus and Linear Algebra : ME Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.
3. VTU Handbook is permitted.*

Module - 1			M	L	C
Q.1	a.	With the usual notation prove that $-\tan \phi = r \frac{d\theta}{dr}$.	6	L2	CO1
	b.	Find the angle between the curves $r = a(1 + \cos\theta)$ and $r = b(1 - \cos\theta)$.	7	L2	CO1
	c.	Find the radius of curvature for the curve $r^n = a^n \cos n\theta$.	7	L2	CO1
OR					
Q.2	a.	Find the angle between the radius vector and tangent to the curve $\frac{2a}{r} = (1 - \cos\theta)$.	6	L2	CO1
	b.	Find the Pedal equation for the curve $r(1 - \cos\theta) = 2a$.	7	L2	CO1
	c.	Find the radius of curvature for the curve $x^3 + y^3 = 3axy$ at the point $(3a/2, 3a/2)$.	7	L2	CO1
Module - 2					
Q.3	a.	Apply Maclaurin's series to prove that $\sqrt{1 + \sin 2x} = 1 + x - \frac{x^2}{2} - \frac{x^3}{6} + \dots$	6	L3	CO1
	b.	Evaluate $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x}\right)^{1/x}$.	7	L2	CO1
	c.	Determine the extreme value of the function $f = x^3 + y^3 - 3x - 12y + 20$.	7	L2	CO1
OR					
Q.4	a.	If $u = x^3 - 3xy^2 + x + e^x \cos y + 1$ show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$	6	L2	CO1
	b.	If $u = F(x-y, y-z, z-x)$ then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$	7	L2	CO1
	c.	If $u = \frac{yz}{x}, v = \frac{zx}{y}, w = \frac{xy}{z}$ Prove that $\frac{\partial(u, v, w)}{\partial(x, y, z)} = 4$	7	L2	CO1

Module - 3						
Q.5	a.	Solve $\frac{dy}{dx} + \frac{y}{x} = y^2 x$	6	L2	CO1	
	b.	Solve the differential equation $y(2xy + 1) dx - xdy = 0$.	7	L2	CO1	
	c.	The number N of bacteria in a culture grew at a rate proportional to N. The value of N was initially 100 and increased to 332 in one hour. What would be the value of N after $1\frac{1}{2}$ hours?	7	L3	CO1	
OR						
Q.6	a.	Solve $r \sin \theta - \cos \theta \frac{dr}{d\theta} = r^2$	6	L2	CO1	
	b.	Apply the concept of differential equations to find the orthogonal trajectories $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1$	7	L3	CO1	
	c.	Solve the differential equation $(x^2 + y^2 + x)dx + xydy = 0$	7	L2	CO1	
Module - 4						
Q.7	a.	Find the Rank of the matrix $A = \begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 4 \end{bmatrix}$	6	L2	CO2	
	b.	Apply Gauss - Seidel method to approximate the solution of the system $20x + y - 2z = 17$, $3x + 20y - z = 18$, $2x - 3y + 20z = 25$ by choosing initial solution (0,0,0). Perform three iterations.	7	L3	CO2	
	c.	Test for consistency and solve $x + 2y + 3z = 14$, $4x + 5y + 7z = 35$, $3x + 3y + 4z = 21$	7	L2	CO2	
OR						
Q.8	a.	Apply Gauss-elimination method to find the solution of system equations $x + y + z = 9$, $x - 2y + 3z = 8$, $2x + y - z = 3$.	6	L3	CO2	
	b.	Apply Jordan method. To find inverse of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$	7	L3	CO2	
	c.	Apply LU decomposition method to find the solution of the system $AX = b$, where $A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$, $b = \begin{bmatrix} 8 \\ 14 \end{bmatrix}$	7	L3	CO2	

Module - 5

Q.9	a.	Apply Moore Penrose Pseudo inverse method to find the inverse of the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & -1 \end{bmatrix}$	6	L3	CO2
	b.	Reduce the matrix $A = \begin{bmatrix} -2 & 5 \\ -3 & 6 \end{bmatrix}$ into diagonal form.	7	L2	CO2
	c.	Apply Rayleigh power method to find the largest eigen value and the corresponding eigen vector of the matrix $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ by taking the initial eigen vector as $[1 \ 1 \ 1]'$	7	L3	CO2
OR					
Q.10	a.	Apply Cayley Hamilton theorem to find the inverse of the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$	6	L3	CO2
	b.	Determine the eigen values and the corresponding eigen vectors for the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$	7	L2	CO2
	c.	Find the characteristic and minimal polynomial of the block matrix $A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$	7	L2	CO2

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First Semester B.E/B.Tech. Degree Examination, June/July 2026
Differential Calculus and Linear Algebra: ME Stream

Time: 3 hrs.

Max. Marks:100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks , L: Bloom's level , C: Course outcomes.
 3. VTU Formula Hand Book is permitted.

Module - 1			M	L	C
1	a.	Derive the angle between radius vector and tangent of the polar curve $r = f(\theta)$	6	L2	CO1
	b.	Find the angle of intersection of the curves $r^2 \sin 2\theta = 4$ and $r^2 = 16 \sin 2\theta$	7	L2	CO1
	c.	Find the radius of curvature of the curve $y^2 = \frac{a^2(a-x)}{x}$, where the curve meets X-axis.	7	L2	CO1
OR					
2	a.	Find the pedal equation for the curve $r^2 = a^2 \sec 2\theta$	6	L2	CO1
	b.	Find the angle of intersection of the curves $r = \frac{a}{1+\cos \theta}$ and $r = \frac{b}{1-\cos \theta}$	7	L2	CO1
	c.	Find the radius of curvature of the curve $x = a(\cos t + t \sin t)$ and $y = a(\sin t - t \cos t)$	7	L2	CO1
Module - 2					
3	a.	Using Maclaurin's series expand $\log(1+e^x)$ up to the term containing x^4	6	L3	CO1
	b.	Apply L'Hospital's rule to solve $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x + d^x}{4} \right)^{1/x}$ and $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^2}$	7	L2	CO1
	c.	If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$	7	L2	CO1

Investigating
x + y + z
Unique

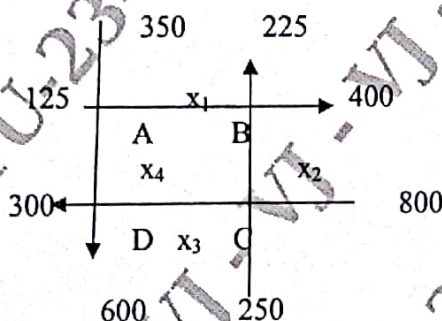
OR				
4	a.	If $u = u\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$ then prove that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$.	6	L1
	b.	If $u = x^2 + y^2 + z^2$, $v = xy + yz + zx$, $w = x + y + z$ find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$	7	L2 CO1
	c.	Find the extreme values of the function $f(x,y) = x^3 + y^3 - 3axy$	7	L2 CO1
Module - 3				
5	a.	Solve $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$	6	L2 CO1
	b.	Analyse the differential equation and solve $x^3 \frac{dy}{dx} - x^2 y = -y^4 \cos x$	7	L2 CO1
	c.	If 30% of a radioactive substance disappears in 10 days, how long will it take for 90% of it to disappear?	7	L3 CO1
OR				
6	a.	Solve the differential equation $(x^2 + y^3 + x) dx + xy dy = 0$	6	L2 CO1
	b.	Find the orthogonal trajectories of the curve $r^n \cos \theta = a^n$	7	L3 CO1
	c.	A bacterial culture, growing exponentially, increases from 100 to 400 gms in 10hrs. How much was present after 3 hrs, from the initial instant?	7	L2 CO1
Module - 4				
7	a.	Find the rank of the matrix $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$	6	L2 CO2
	b.	Apply Gauss seidel method to approximate the solution of the system $2x - 3y + 20z = 25$; $20x + y - 2z = 17$; $3x + 20y - z = -18$ by choosing initial solution (0,0,0). Carryout 4 iterations	7	L3 CO2

c.	Investigate the values of λ and μ such that the system of equations $x + y + z = 6$; $x + 2y + 3z = 10$; $x + 2y + \lambda z = \mu$ may have i) Unique solution (ii) Infinite solution (iii) No solution.	7	L2	CO2
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OR

8	a.	Apply Jordan method, to find inverse of matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$	6	L3	CO2
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b.	Write the system of linear equations of the traffic flow in the net of one-way street directions as shown in the figure and find its solution	7	L3	CO2
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c.	Apply LU decomposition method to find the solution of the system $AX = b$, where $A = \begin{bmatrix} 2 & 3 \\ 4 & 9 \end{bmatrix}$, $b = \begin{bmatrix} 13 \\ 33 \end{bmatrix}$	7	L3	CO2
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Module – 5

9	a.	Apply Moore Penrose Pseudo inverse method to find the inverse of the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$	6	L3	CO2
	b.	Reduce the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ into diagonal form.	7	L2	CO2
	c.	Using power method, find the largest eigen value and corresponding eigen vector of the matrix $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ take $[1, 1, 1]^T$ as the eigen vector and perform 5 iterations.	7	L3	CO2

OR				
10	a.	Apply Cayley Hamilton Theorem to find the inverse of the matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$	6	L3 CO2
	b.	Determine the eigenvalues and corresponding eigenvectors for the matrix $A = \begin{bmatrix} 5 & 2 & 0 \\ 2 & 5 & 0 \\ -3 & 4 & 6 \end{bmatrix}$	7	L2 CO2
	c.	Find characteristic and minimal polynomial of the block matrix $A = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 1 & 4 & 0 & 0 \\ 0 & 0 & -4 & 0 \\ 0 & 0 & 1 & -4 \end{bmatrix}$	7	L2 CO2
